

LIME

Local Interpretable
Model-agnostic Explanations

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SCC5819 Topics in Artificial Intelligence · ICMC-USP

Accurate is not the same as accurate for the right reasons.

Marco Ribeiro, who wrote LIME, tells this on himself. A model of his scored well in cross-validation and fell apart in user testing. The cause, after a long debugging: it had learned to detect *how* the positive and negative data were collected — a shortcut that did not exist in the real task.

THE PART THAT STINGS

**He was the expert,
and he could not tell.**

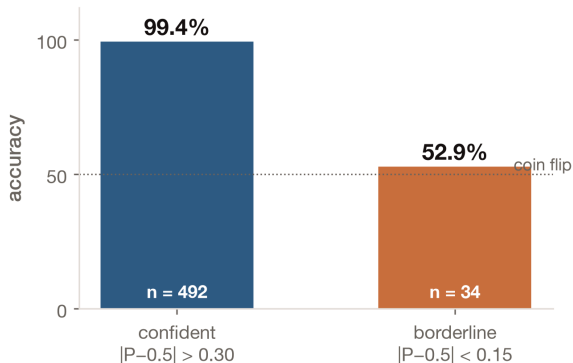
IN HIS OWN WORDS

"Here I was, a PhD student, supposed to be an expert — and it took me the longest time to understand how my model was making predictions."

Marco Túlio Ribeiro · author of LIME

The same shape of problem, in our data.

95.3% accuracy under ten-fold cross-validation — until you split it by confidence. The 6% of patients where the model hesitates are exactly the ones a clinician would send for a second opinion.



Four words, four commitments.

The first commitment is the one people skip: LIME explains one prediction at a time, never the whole model. Explaining everything at once forces you to hide too much. So it builds a simple, readable approximation of how the model behaves around a single instance — without needing to know what kind of model it is.

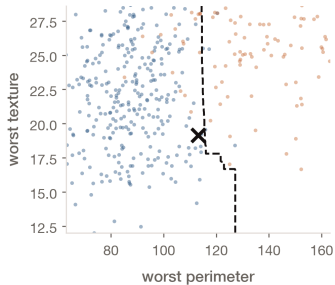
LOCAL**faithful near one point****INTERPRETABLE****readable by a human****MODEL-AGNOSTIC****any model, any data****EXPLANATIONS****input → prediction, legible**

One equation, three demands.

$$\xi(x) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

- 01 Be faithful to the black box — loss $\mathcal{L}$ against f , a RandomForest of 300 trees
- 02 Be faithful *here* — kernel $\pi_x = \sqrt{\exp(-d^2/\nu^2)}$, width $\nu = 0.75\sqrt{30} \approx 4.11$
- 03 Stay readable — penalty Ω : 8 of 30 features, and g is a Ridge with $\alpha = 1$

THE MODEL



● benign ● malignant × patient 67

Dashed: the forest's own boundary in this slice.

THE CODE

```
# the only thing we  
# ever ask of f  
p = model.predict_proba(X)[:, 1]
```

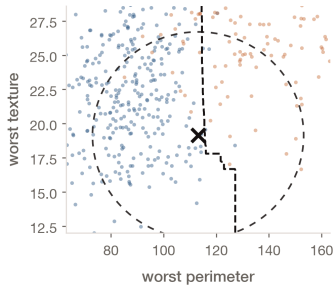
THE IDEA

Patient #67, on the boundary.

Chosen for it: a confident patient pushes the fit's $P=0.5$ contour off the figure entirely. The axes are her rank-1 and rank-3 features — rank 2 correlates 0.98 with rank 1 and would flatten this to a ribbon.

We never look inside f.

THE MODEL



$\pm 1.2\sigma$ reference — not LIME's kernel.

THE CODE

```
# one width per feature,  
# on its own scale  
nu = 0.75 * np.sqrt(30)
```

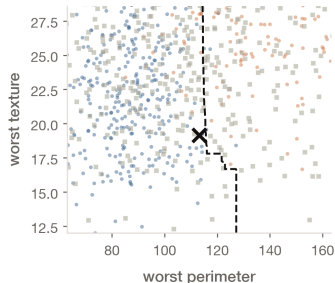
THE IDEA

There is no single radius.

Each feature has its own scale. But note the real kernel is far wider than this frame: the median neighbor sits $\sim 6\sigma$ away and still carries a third of the weight.

Remember that width. It returns twice.

THE MODEL



Squares: synthetic neighbors, all 30 features resampled.

THE CODE

```
# centred on the DATASET,  
# not on the patient  
Z = mu + sigma * rng.normal(  
    0, 1, size=(n, 30))
```

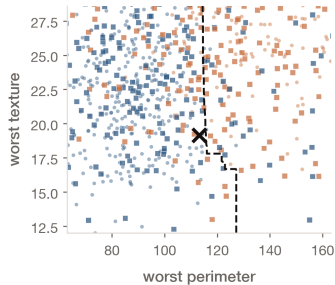
THE IDEA

These are not possible patients.

Sampling each feature independently breaks the correlations that make a tumour measurable: ~76% carry a negative measurement.

This comes back in the limitations.

THE MODEL



Colour: what f predicts for each neighbor.

THE CODE

```
# the one moment the real  
# model is queried  
fz = model.predict_proba(Z)[: , 1]
```

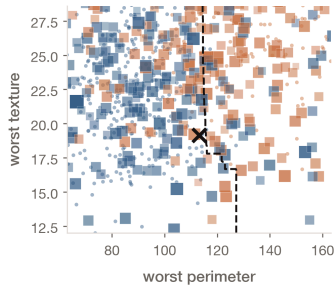
THE IDEA

They do not split along the line.

68% agreement against a 53% baseline.
Each square has random values on the other 28 features, while the contour freezes them at hers.

A 2-D slice cannot decide a 30-D prediction.

THE MODEL



Size and opacity encode the weight.

THE CODE

```
d = norm(Zs - x_scaled, axis=1)
w = np.sqrt(
    np.exp(-d**2 / nu**2))
```

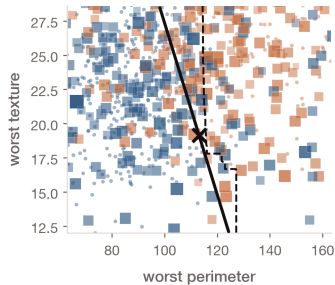
THE IDEA

Closer neighbors count more.

Distance is over all 30 standardized features, not the two on screen — a square that looks far can still be heavy.

Look how little the weights actually vary.

THE MODEL



Solid: the fit, drawn through the patient.

THE CODE

```
Ridge(alpha=1).fit(  
    Zs[:, top8], fz,  
    sample_weight=w)
```

THE IDEA

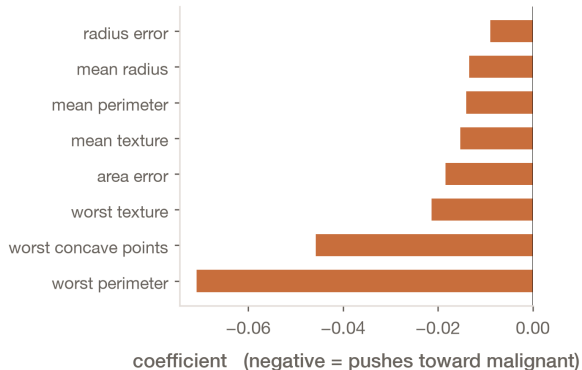
One straight line for a staircase.

Fitted on the 8 selected features, weighted by π . Drawn through the patient rather than at $P=0.5$ — the next section explains why.

$R^2 = 0.36$

What the library actually hands you.

Not a line — a list of coefficients. Every one of them is negative here: all eight selected measurements push toward malignant, and the model still returns benign. The remaining 22 features and the forest's nonlinearity carry the verdict, and a local linear summary structurally cannot show you that.



A linear fit gives you two things. They are not equally reliable.

DO NOT TRUST

The *level*: $g(x) = 0.499$ while $f(x) = 0.581$.



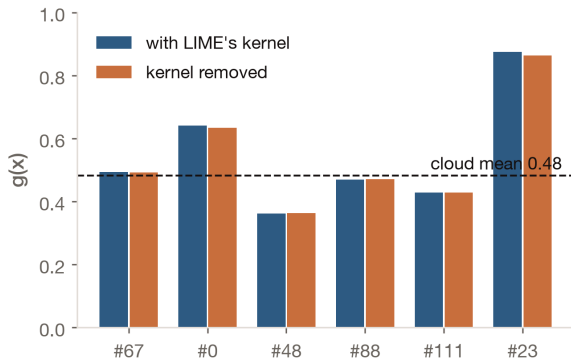
TRUST

The *direction*: 4° from where the model actually changes.

COSINE 0.997 IN-PLANE · 0.57 ACROSS 30-D

The level is biased, and here is why.

Delete the proximity kernel entirely and almost nothing changes — $g(x)$ moves by ~ 0.002 , the coefficient directions stay aligned at cosine 0.9999. So the fit is effectively global. A global fit explaining a third of the variance shrinks its predictions toward the cloud's mean of 0.48, which is why $g(x)$ lands near 0.5 for almost any patient. The intercept describes the cloud, not her.

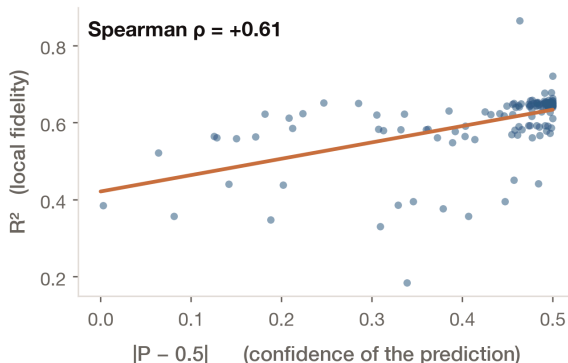


A high R^2 can certify that nothing is happening.

R^2 tracks how *confident* the prediction was, not how good the explanation is: $\rho = +0.61$. The intuitive reason — the model saturates far from the boundary, so a line fits it easily — is false, three times over:

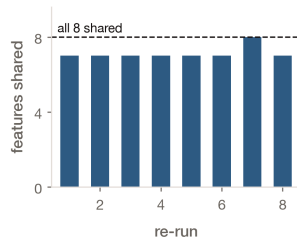
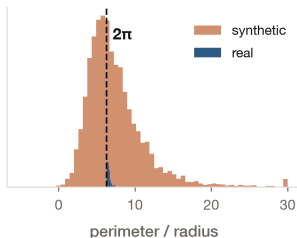
- 01 It never saturates where LIME samples: 2 of 5,000 draws, against 71% of real patients.
- 02 Weighted variance of f is identical in both groups — they share one cloud.
- 03 $R^2 = 1 - \text{SSE}/\text{SST}$, so a flatter target makes high R^2 *harder*, not easier.

The replacement story fails too. The correlation is robust and we cannot say why.



LIME is useful. It is not neutral.

- 01 Samples impossible data
- 02 The kernel barely localizes
- 03 R^2 measures the wrong thing
- 04 The explanation wobbles



OFF-MANIFOLD perimeter / radius

INSTABILITY top-8 across re-runs

Three of the limitations Molnar lists, measured on our own run.

Neighborhood definition, sampling that ignores feature correlation, and instability across runs.

Which number would you report to the board?

Not 95.3% alone. 95.3% together with: on the 6% of patients where the model hesitates it is right about half the time — and here is a tool that tells you which patient you are looking at, along with how much of what it says you should believe.

KEY TAKEAWAY

LIME's value is not that it is never wrong. It is that, once you measure it, you can say precisely how it is wrong.

Thanks!



github.com/wbendinelli/interpretable-ml-lectures

Notebooks, figures and these slides. Every number in this deck is printed by the notebooks in module 03.